CarbonTrace: Human-Reviewed AI for Traceable Material Records and Emission Scenario Estimates

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Project demonstration: https://carbontrace.raynaamara.com
Video project: https://www.youtube.com/watch?v=gCZY3bU5FSc

2. ABSTRACT

Comparing landfills and recycling requires material records, weights, and emission references that can be checked. Missing or incorrect inputs make environmental estimates difficult to interpret. This project developed CarbonTrace (https://carbontrace.raynaamara.com), a browser-based AI-assisted system for PET (Polyethylene Terephthalate) bottles, cardboard, and paper-faced drywall. A classifier using frozen MobileNetV1 features suggests the material; users confirm the category and enter a weight. Saved records keep the AI suggestion, user decision, and emission-factor snapshot separate. One dataset contains 120 labeled photos, with 40 per class, divided into 78 training photos and 42 evaluation photos. Training augmentation produced 312 examples. All 42 evaluation predictions matched their supplied labels. All 146 software tests passed, including record-validation and factor-snapshot checks. Scenario calculations used WARM (Waste Reduction Model) and PLACES (Plastic Lifecycle Assessment Calculator for the Environment and Society) references and a 1 kg analytical unit, without measured waste collections. The lower-emission scenario depended on the material and reference assumptions; a lower GHG (Greenhouse Gas) estimate does not establish disposal safety. The prototype demonstrates a traceable workflow for reviewing material records and scenario estimates. It may support environmental education and early comparisons. Independent evaluation, measured weights, and treatment evidence remain necessary before claiming actual emission reductions.

3. INTRODUCTION

3.1. The environmental-record problem

Comparing landfill and recycling requires records that connect a material and its weight to a suitable emission reference. When these inputs are missing, incorrectly assigned, or separated from their sources, the resulting estimate becomes difficult to check. For example, using a factor for the wrong material can change the comparison even when the weight and arithmetic are correct. This creates a record-keeping problem for anyone trying to explain the environmental meaning of a calculation.

Image classification can help by suggesting a material category from a photo. However, a photo does not supply a measured weight or verify material composition, contamination, or safety. Users still need to check the material and enter a weight. The record must also identify the scenarios being compared, without assuming that either scenario describes what actually happened to the waste.

CarbonTrace covers PET bottles, cardboard, and paper-faced drywall. PET means polyethylene terephthalate, a plastic commonly used for drink bottles. Drywall consists of a gypsum core with paper facing; this category does not cover all gypsum products or construction debris. The EPA (Environmental Protection Agency) drywall example also shows why a lower greenhouse gas estimate alone cannot establish the safest environmental choice [10].

An emission factor expresses the modelled greenhouse gas impact per unit mass of a stated material under a defined scenario. EPA's WARM model provides screening-level comparisons of waste-management options. It is not intended for annual greenhouse gas inventories or final site-specific decisions [5]. PLACES provides plastic-waste calculations with regional assumptions, including Indonesia [6]. CarbonTrace uses selected Indonesia/PET scenarios from PLACES. Differences in material coverage and calculation assumptions mean that the source and scope must accompany each comparison.

CarbonTrace addresses the recording task by keeping the AI suggestion, user-confirmed category, entered weight, and emission-factor snapshot distinct within each saved record. This allows a reviewer to trace an estimate to its inputs and source assumptions. The prototype is

intended for environmental learning and early scenario comparisons. A saved record does not verify actual recycling, demonstrate emission reductions caused by the project, or establish carbon credits.

3.2. Related work and design contribution

Image-based waste classification has been explored using datasets such as TrashNet. Thung and Yang collected 2,527 images across six classes: glass, paper, cardboard, plastic, metal, and trash [1]. CarbonTrace uses a separate dataset covering PET bottles, cardboard, and paper-faced drywall. TrashNet was not used for training or evaluation. Differences in images, class definitions, and evaluation procedures prevent a direct comparison of reported accuracy.

MobileNet provides a computationally efficient architecture for image recognition through depth wise separable convolutions [2]. CarbonTrace uses frozen, pretrained MobileNetV1 features and trains a final classifier for its three material classes. TensorFlow.js supports machine-learning computation in web browsers and Node.js [3]. In the application, inference means applying the trained model to a photo. This takes place on the user's device without sending the photo to an external prediction service.

WARM and PLACES provide documented references for comparing waste-management scenarios [5,6]. These references supply the emission factors, while the image classifier supplies a material suggestion. Users confirm the material and enter a weight before a record is saved. Keeping these roles separate makes clear which inputs come from the model, the user, and the published references.

CarbonTrace's design contribution is an implemented workflow connecting local image classification with human confirmation and traceable scenario calculations. Each saved record preserves the AI suggestion separately from the user's decision and retains the factors, source, and calculation assumptions used. Later changes to factor settings do not silently rewrite earlier records. The study evaluates internal classification results and the software behaviour of this workflow. It uses an existing neural-network architecture and does not establish comparative superiority over other recycling applications.

4. GOAL / RESEARCH QUESTION

The goal was to develop and evaluate an AI-assisted workflow for creating traceable material records and comparing waste-management emission scenarios. The evaluation examines image classification, preservation of recorded inputs and sources, and the effect of material and reference choices on calculated results.

The study addresses three questions:

- **RQ1:** How often does the three-class model match the supplied labels in the 42-photo evaluation set, overall and by material class?
- **RQ2:** Do saved records preserve the AI suggestion separately from user-confirmed inputs and retain their factor and source snapshots when factor settings change?
- **RQ3:** For a 1 kg analytical mass, how does the estimated difference between baseline and recycling scenarios vary across supported materials and documented factor sets?

The evidence consists of supplied photos, saved model files, automated software-test results, and published emission factors. The scenario comparisons use an analytical mass rather than measured waste collections. Improvements in user accuracy or recording speed, actual waste treatment, and real emission reductions were not measured.

5. AI MODEL / SYSTEM

5.1. System organization

CarbonTrace is a browser-based system written in TypeScript. Shared hosting delivers the application files, catalog photos, and trained model files. TensorFlow.js 4.22.0 performs image classification on the user's device; photos are not sent to an external prediction service. The model evaluated in this study is ct-mobilenet-913cfe5aa706, trained on 26 October 2026 [8].

The recording workflow begins with a catalogue photo or a locally selected image. Users may request an AI suggestion or continue without AI. They then confirm a material category and enter a positive weight in grams. An optional calculation compares documented waste-management scenarios. A record is stored only after the user explicitly saves it.

Records and factor preferences are stored in local storage. Prepared copies of uploaded photos are stored in IndexedDB. These are browser-local storage mechanisms, not a shared database that synchronizes records across devices. Table 1 summarizes the system components and their boundaries.

Table 1. System components, their functions, and operational boundaries. Source: project code, model report, and automated test records [8,9].

Component	Function	Boundary
Image handling	Accepts a catalog photo or a local JPG/PNG/WebP image.	A photo does not provide a measured weight or verify material composition.
AI classifier	Optionally produces PET, cardboard, and drywall scores and a suggested category.	No automatic rejection of unsupported materials or hazard detection.
Human confirmation and recording	Requires an explicit material choice and a positive weight in grams before saving.	Input validation does not independently verify the material or physical weight.
Scenario calculator	Optionally calculates baseline and alternative estimates using a documented, material-specific factor pair.	Unsupported factors remain unavailable, not zero. Calculated estimates do not verify actual treatment.
Provenance and storage	Saves records, applicable AI metadata, factor snapshots, and uploaded-photo copies locally.	Records are browser-local, without shared-account synchronization or certified verification.

Provenance means retaining the information needed to trace a result to its inputs and calculation basis. CarbonTrace links each record to its photo reference, user-confirmed material, and entered weight. AI-assisted records also retain the model and analysis metadata. Records with emission estimates preserve the factors, source, and assumptions used. These details support review of the calculation without independently verifying the physical material, weight, or treatment outcome.

5.2. Dataset and evaluation units

The study uses one dataset of 120 labelled photographs covering PET bottles, cardboard, and paper-faced drywall, with 40 photos per class. A reference label is the material category assigned to a photo before evaluation. According to the dataset declaration, the photographs show objects

with different physical objects across IDs. Material identity and object independence were not independently audited, so the evaluation measures agreement with the supplied labels.

The dataset contains 78 training photos and 42 evaluation photos. Each class contributes 26 training photos and 14 evaluation photos, as summarized in Table 2. Evaluation photos were excluded from fitting the classifier head.

Table 2. Source-photo roles in the current three-class model. Exact IDs and hashes are in the report [8].

Role	Use	PET	Cardboard	Drywall	Total
Training	Used to fit the classifier head	26	26	26	78
Evaluation	Kept out of fitting; used to check predictions	14	14	14	42
Dataset total	IDs 001-040 per material	40	40	40	120

The training and evaluation assignments were fixed before fitting the current classifier. The first 26 were assigned to training and the remaining 14 to evaluation. The application catalog also includes 12 exploratory CT photos outside the formal dataset. It therefore contains 132 photos in total. The exploratory photos are excluded from the formal classification metrics and confusion matrix.

5.3. Image processing and trained model

The original PNG files are retained in the project archive. Catalog display, training, and formal evaluation use prepared JPEG copies with a longest side of at most 1,024 pixels and JPEG quality setting 85. For user uploads, the application checks the file format, size, and image dimensions before creating a local JPEG copy. Uploaded copies also have a longest side of at most 1,024 pixels and use the browser's JPEG quality parameter of 0.90. Reusing the saved catalog files keeps the encoded evaluation inputs fixed. However, JPEG decoding can differ between browsers, so identical files do not guarantee identical RGB values or scores across environments.

In the browser, the prepared image is drawn onto a canvas at its natural pixel dimensions. The displayed preview size therefore does not determine the pixels supplied for analysis. The RGB image is then resized directly to 224 by 224 pixels using bilinear interpolation with aligned corners.

The model uses transfer learning with an ImageNet-pretrained MobileNetV1 feature extractor and a width multiplier of 0.25. Features are taken from layer conv_pw_13_relu. Global average pooling averages each feature map across its spatial dimensions. The resulting 256-value vector is normalized by its Euclidean length, with a small numerical stabilizer [2,3,8].

The feature extractor remains frozen during training. Only the final dense classifier, called the head, is trained. It contains 256 x 3 weights and 3 biases, giving 771 trainable parameters. Its three softmax outputs correspond to PET, cardboard, and drywall and sum to one. These weights and biases are model parameters, not material weights in grams. Table 3 lists the model and training configuration.

Each of the 78 training photos supplies four variants: the original, a horizontal flip, and brightness multipliers of 0.9 and 1.1. Brightness-adjusted pixel values are clipped to the range 0–255. Feature extraction produces 312 training vectors from these variants. Augmentation introduces no additional physical objects, and evaluation photos are excluded from this process.

Table 3. Fixed training settings. Source: training script and frozen report [8,9].

Setting	Value
Feature extractor	Frozen ImageNet-pretrained MobileNetV1 0.25
Input / feature size	224 x 224 RGB / 256 normalized features
Learned head	Dense layer, 3 softmax outputs
Trainable parameters	256 x 3 weights + 3 biases = 771
Source training images	78
Training variants	Original, horizontal flip, brightness x0.9, brightness x1.1
Training feature vectors	312; not 312 independent objects
Optimizer / loss	Adam / categorical cross-entropy
Learning rate / kernel L2	0.03 / 0.001
Epochs / batch / shuffle	200 / 312 / false
Initializer seed	20261024
Separate validation set	None

The head is trained for 200 epochs using all 312 feature vectors in each batch, with shuffling disabled. An epoch is one complete pass through the training examples. Adam optimizes categorical cross-entropy with a learning rate of 0.03. Kernel L2 regularization, with coefficient 0.001, discourages large classifier weights. The weight initializer uses seed 20261024. These settings were retained from the previous model without tuning against the current evaluation results. No separate validation set was used [8,9].

The displayed photo score is the softmax output for the selected category. Calibration examines how close prediction scores correspond to observed correctness [4]. Calibration was not evaluated in this study, so the score should not be interpreted as a verified probability of correct material identification. Predictions are computed from image pixels; filenames identify the photos but do not determine their predicted categories.

5.4. Evaluation measures and baseline

Evaluation uses 42 labelled photos excluded from fitting the current classifier head, with 14 photos per material class. The study examines predictions archived at the 26 October 2026 model update and checks their repeatability under the specified model, image-processing, and decoding conditions [8,9]. Repeating a prediction checks computational consistency; it does not add an independent evaluation case.

For each photo, the model selects the category with the highest softmax score. A prediction is counted as correct when it matches the supplied reference label. Accuracy is the number of these matches divided by the number of evaluation photos. These metrics assess the model's predictions before any user correction.

Precision measures how many photos predicted as a class have that reference label. Recall measures how many photos with that reference label are correctly identified. F1 combines precision and recall using their harmonic mean. Macro-F1 is the unweighted average of the three class-specific F1 scores, giving each material class equal weight. The confusion matrix records count for each combination of reference label and predicted category.

A constant-class rule that always predicts PET provides a simple baseline. It correctly identifies the 14 PET-labelled photos and misses the 28 cardboard and drywall photos, giving 14/42 matches, or 33.3% accuracy. Because the evaluation set is balanced, always predicting either of the other classes would give the same accuracy. This baseline uses the same 42 photos as the trained classifier and provides a basic comparison without training another model.

5.5. Scenario calculation and input provenance

CarbonTrace compares a baseline scenario, such as landfill, with an alternative, such as recycling, using the same material mass. The calculation uses the user-confirmed material, entered weight, and a documented factor pair. The AI classifier supplies a material suggestion; it does not measure weight or predict emission factors.

The scenario difference is calculated as:

Delta E = (mass in grams / 1000) x (EF_baseline − EF_alternative).

Each emission factor has units of kg CO2e per kg of material. Delta E has units of kg CO2e for the entered mass. CO2e expresses different greenhouse gases on a common scale. A positive difference means the alternative has a lower modelled impact; a negative difference means the baseline is lower. Zero means equal modelled impacts, not zero emissions.

Negative factors can include avoided virgin-material production or modelled carbon storage. They do not represent direct CO2 removal by the photographed item. Their interpretation depends on the processes and effects included within the source's calculation boundary [5,6,10].

WARM supplies Sets 1, 2, and 5. Sets 1 and 2 cover PET and Corrugated Containers under US assumptions. Set 1 uses US-average landfill conditions; Set 2 assumes no landfill-gas collection or recovery. Set 5 uses the Drywall factors from Summary Data cells E53/C53 for regular paper-faced gypsum board. It should not be applied as a general factor for cement board, pure gypsum, or mixed or contaminated construction debris. These US references do not establish conditions at an Indonesian facility [5,10].

WARM factors are converted from metric tonnes CO2e per US short ton of material to kg CO2e/kg:

EF_application = EF_WARM x 1000 / 907.18474.

PLACES supplies Sets 3 and 4 for Indonesia/PET, with Sanitary Landfill and Open Dump baselines respectively. The saved queries used 1,000,000 metric tonnes of PET and reported impacts in tonnes CO2e. Dividing each impact by the input quantity gives tonnes CO2e per tonne PET, numerically equivalent to kg CO2e/kg PET. The large query quantity reduces the effect of display rounding; it is not an observed project mass. The WARM short-ton conversion is not applied. Source queries are listed in Reference 6 and may require sign-in [6].

Normalized preset factors are rounded to six decimal places before use. Scenario values and differences are calculated from these stored coefficients, not from rounded display values. The decimal places do not indicate measurement precision. Unsupported material–factor combinations remain unavailable rather than receiving zero or another material's factor.

When estimates are enabled, material confirmation applies WARM Set 1 for PET/cardboard or Set 5 for drywall only if that material has no configured factors. Existing choices are preserved. These defaults do not verify local conditions. Users may disable the estimate and save the material and weight without a factor snapshot or CO2e result, while retaining their factor preferences.

For records with estimates, provenance includes a snapshot of the factor values, scenario names, source title and URL, scope, parameter-set identity, and factor date. Applying another reference changes the selected material's preferences without automatically rewriting saved records. Updating an earlier record requires an explicit edit and save.

Numerical scenario comparisons in this paper use a 1 kg analytical reference mass, not measured waste collections. The companion 132-photo catalogue workbook contains illustrative masses for demonstrations; its measured-mass cells remain empty. These values were not used for image training, classification evaluation, or physical-measurement claims. Actual use requires a suitable scale and confirmation that the weighed material falls within the chosen factor's scope.

6. RESULTS AND EVALUATION

6.1. Image-level and class-level results

Table 4 lists the predictions for the 42-photo evaluation portion of the single 120-photo dataset. The evaluation set contains 14 photos from each class: PET bottles, cardboard, and paper-faced drywall. The same trained model and preprocessing pipeline were applied to all evaluation photos. “Top softmax” is the score assigned to the predicted category. Table values are rounded for presentation, while numerical checks use the stored, unrounded scores [8].

Table 4. Saved predictions for all 42 evaluation photos, ordered by material class and image ID. Source: model report [8].

Image ID	Reference	Predicted	Top softmax
SYN-PET-009	PET	PET	0.948544
SYN-PET-010	PET	PET	0.942254
SYN-PET-019	PET	PET	0.777735
SYN-PET-020	PET	PET	0.870737
SYN-PET-022	PET	PET	0.883935
SYN-PET-023	PET	PET	0.960665
SYN-PET-028	PET	PET	0.762395
SYN-PET-030	PET	PET	0.780055
SYN-PET-031	PET	PET	0.712604
SYN-PET-032	PET	PET	0.759680
SYN-PET-033	PET	PET	0.805880
SYN-PET-034	PET	PET	0.872351
SYN-PET-036	PET	PET	0.725331
SYN-PET-040	PET	PET	0.975313
SYN-CRD-009	Cardboard	Cardboard	0.890216
SYN-CRD-010	Cardboard	Cardboard	0.931918
SYN-CRD-019	Cardboard	Cardboard	0.731068
SYN-CRD-020	Cardboard	Cardboard	0.676077
SYN-CRD-021	Cardboard	Cardboard	0.909556

Image ID	Reference	Predicted	Top softmax
SYN-CRD-022	Cardboard	Cardboard	0.935923
SYN-CRD-023	Cardboard	Cardboard	0.927207
SYN-CRD-024	Cardboard	Cardboard	0.948877
SYN-CRD-028	Cardboard	Cardboard	0.810071
SYN-CRD-029	Cardboard	Cardboard	0.907089
SYN-CRD-030	Cardboard	Cardboard	0.698505
SYN-CRD-031	Cardboard	Cardboard	0.758114
SYN-CRD-034	Cardboard	Cardboard	0.670567
SYN-CRD-036	Cardboard	Cardboard	0.854734
SYN-DRYWALL-008	Drywall	Drywall	0.883112
SYN-DRYWALL-014	Drywall	Drywall	0.883057
SYN-DRYWALL-016	Drywall	Drywall	0.902950
SYN-DRYWALL-021	Drywall	Drywall	0.863961
SYN-DRYWALL-022	Drywall	Drywall	0.843848
SYN-DRYWALL-023	Drywall	Drywall	0.896841
SYN-DRYWALL-024	Drywall	Drywall	0.895368
SYN-DRYWALL-025	Drywall	Drywall	0.756575
SYN-DRYWALL-027	Drywall	Drywall	0.852069
SYN-DRYWALL-029	Drywall	Drywall	0.877282
SYN-DRYWALL-030	Drywall	Drywall	0.898996
SYN-DRYWALL-031	Drywall	Drywall	0.880475
SYN-DRYWALL-038	Drywall	Drywall	0.779836
SYN-DRYWALL-040	Drywall	Drywall	0.723907

Table 5. Class-level and overall results for the 42-photo evaluation set. Overall precision, recall, and F1 are macro averages. Source: Table 4 and the model report [8].

Class	Photos	Matches	Precision	Recall	F1
PET	14	14/14	1.00	1.00	1.00
Cardboard	14	14/14	1.00	1.00	1.00
Drywall	14	14/14	1.00	1.00	1.00
Overall (macro)	42	42/42	1.00	1.00	1.00

All 42 model predictions matched their supplied reference labels, giving an overall accuracy of 1.00. The confusion matrix contains 14 correct predictions for each class and zero off-diagonal counts. Precision, recall, and F1 are therefore 1.00 for each class, and macro-F1 is 1.00, as summarized in Table 5. These metrics describe label agreement within this internal evaluation set and do not establish general real-world accuracy.

Across all 42 evaluation photos, top softmax scores ranged from 0.670567 to 0.975313. Scores varied between individual images even though every predicted category matched its reference label. These outputs indicate the model's relative preference among the three supported classes; they are not calibrated probabilities of correct physical material identification.

The archived results were obtained using the saved model, prepared catalogue JPEG files, and documented decoding and preprocessing conditions. Browser JPEG decoding can produce slightly different RGB values and scores. Re-encoding a photo can also change its pixels, so an uploaded copy of the same photo need not receive exactly the archived catalogue score. Repeating inference or creating another encoded copy of the same photo does not add an independent evaluation case.

6.2. Confusion matrix and evaluation limits

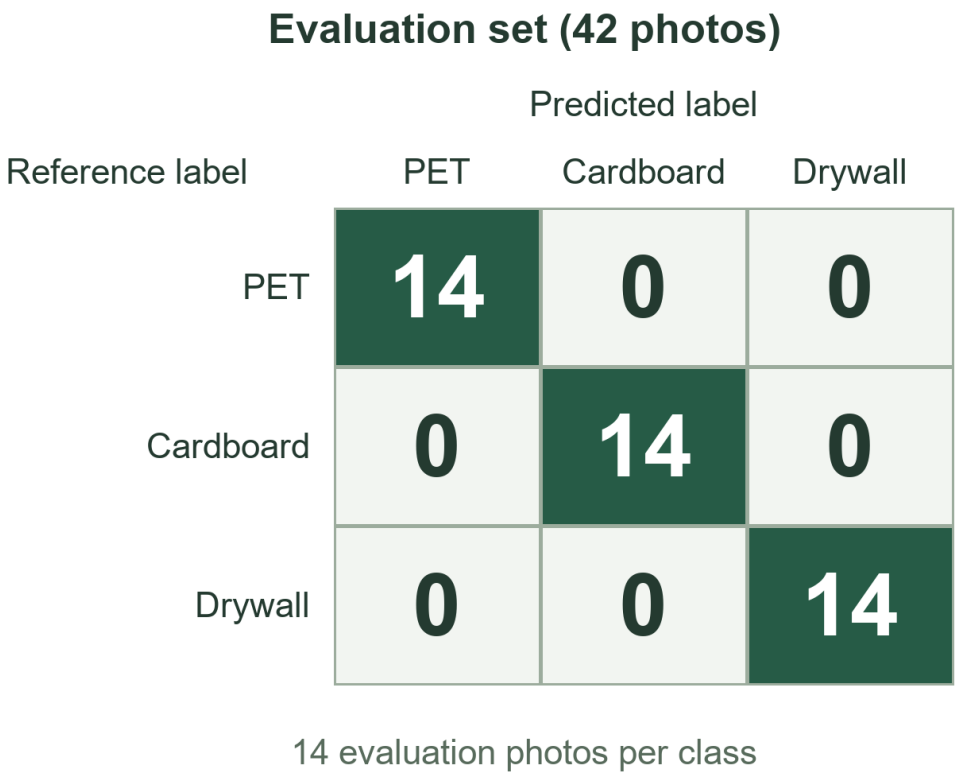


Figure 1. Confusion matrix for the 42-photo evaluation set, with 14 photos per class. Rows are supplied reference labels; columns are model predictions. Source: saved report [8].

Figure 1 summarizes the classifier's results on the evaluation portion of the single 120-photo dataset. It contains 14 correct predictions for each material class and no off-diagonal counts. The matrix describes agreement with the supplied labels under the documented evaluation conditions.

On the same balanced evaluation set, a constant-class rule that always predicts PET produces 14/42 matches, or 33.3% accuracy. Always selecting either of the other classes gives the same accuracy. The trained model's 42/42 result exceeds this basic baseline on these images. No comparison with another trained model or a person was performed.

All 42 evaluation photos were excluded from fitting the current classifier head. However, 28 had already appeared in earlier model evaluations, so this set is not a wholly new independent benchmark. Earlier model files remain available for version checks [8]. Further independent evaluation across all three classes is needed.

Human confirmation is required before saving a record. Further evaluation should use independently checked objects, varied backgrounds and lighting, and fresh examples across all three classes.

6.3. Software verification and record traceability

In the archived verification run on 26 October 2026, the full software test suite was executed. All 146 tests passed, with none failed, skipped, or unfinished. These tests checked software behavior and data consistency; they do not represent additional photographs or participants [9]. Table 6 summarizes their coverage.

Table 6. What the software tests cover. The rows group types of checks. Source: test output and test files [9].

Area	Interpretation
Model/data artifacts	Check the frozen artifact chain.
Catalog images	Does not authenticate physical objects.
Record logic	Preserves declared input rules.
Factor handling	Rejects invalid or unsupported factor assignments.
Historical data	Current settings do not silently rewrite history.
Uploads and exports	Does not validate a real collection mass.
Interface contracts	Not independent usability or full accessibility study.

Browser checks loaded all 132 catalogue photos, comprising the 120-photo formal dataset and 12 exploratory application examples. A viewport-repeatability check performed six inference runs using two existing photos at three screen widths. Within the tested browser, each photo retained the same decoded-RGB digest and prediction scores across those widths. A separate set of 350 layout checks covered language, theme, and application-view combinations. These checks did not increase the size of the formal 42-photo classification evaluation.

Record checks confirmed that an AI suggestion remains distinguishable from the user's confirmed material. Records analysed with the current model retain all three class scores, the model version, analysis time, and the decoded RGB pixels and image dimensions. Uploaded-photo records also retain metadata identifying the prepared JPEG copy. Manual records contain no invented AI prediction when analysis is not used.

Factor-snapshot checks confirmed that saved records retain the factor values, scenario names, source, scope, parameter-set identity, and factor date used for their calculations. Changing factor preferences does not automatically rewrite earlier records; applying different factors to an existing record requires an explicit edit and save. Display-language and colour-theme changes leave the stored records and factors unchanged.

Compatibility checks covered records and settings created by earlier software versions. Their original model identifiers, available score fields, and source values are preserved without inventing missing predictions or factors. History and Overview keep incompatible material/reference calculation bases separate rather than producing a cross-basis CO2e total.

Saving an existing sample ID updates its record instead of creating a duplicate, an operation called upsert. Disabling the optional estimate leaves that record without a factor snapshot or CO2e result while retaining factor preferences. The 32-column CSV exports all active records,

including selected AI and factor-provenance fields, without including photo files. Together, these checks verify implemented recording, calculation, and export rules. They do not authenticate physical objects, validate measured weights, establish user-performance improvements, or verify actual waste treatment.

6.4. Unit-mass scenario results and sensitivity

Table 7 applies the same analytical mass of 1 kg to all supported material–reference combinations for PET, cardboard, and paper-faced drywall. The calculations use published emission factors rather than values learned by the image classifier. They represent modeled scenario comparisons, not measured waste quantities or observed emission reductions.

Table 7. Modeled scenario results for a constant analytical mass of 1 kg. Baseline and alternative emission factors are expressed in kg CO2e/kg material; Delta E is expressed in kg CO2e for the 1 kg mass. Delta E equals baseline minus alternative. Sources: WARM v16 and PLACES Indonesia [5,6,10].

Material	Set / baseline	Baseline EF	Alternative EF	Delta E for 1 kg
PET	1: WARM US average	0.022327	−1.141687	1.164014
PET	2: WARM no LFG	0.022327	−1.141687	1.164014
PET	3: PLACES sanitary landfill	0.095579	−2.497533	2.593112
PET	4: PLACES open dump	0.101700	−2.497533	2.599233
Cardboard	1: WARM US average	0.200213	−3.456117	3.656330
Cardboard	2: WARM no LFG	1.833663	−3.456117	5.289780
Drywall	5: WARM US average	−0.067287	0.028756	−0.096043

Under the US-average WARM references, recycling has a lower modelled impact for PET and cardboard, with baseline-minus-alternative differences of 1.164014 and 3.656330 kg CO2e respectively for 1 kg. For paper-faced drywall, the difference is −0.096043 kg CO2e, meaning the landfill scenario has the lower modelled GHG impact under that reference. Together, these results show that the direction and magnitude of the comparison depend on the material and calculation assumptions [5,10].

Reference choices also affect the results. WARM Sets 1 and 2 give identical PET values, while changing the landfill-gas assumption increases the cardboard difference from 3.656330 to 5.289780 kg CO2e. The selected PLACES Indonesia/PET scenarios give differences of 2.593112 kg CO2e for Sanitary Landfill and 2.599233 kg CO2e for Open Dump. Differences between WARM and PLACES reflect their model boundaries, transport rules, waste scenarios, and virgin-material substitution assumptions as well as geography [5,6].

A hypothetical substitution of cardboard factors for PET factors under Set 1 changes the result by |3.656330 − 1.164014| = 2.492316 kg CO2e for the same 1 kg mass. This illustrates the numerical consequence of an incorrect material–factor assignment. It was not an error observed in the image evaluation or a measured improvement attributable to human review.

Negative net factors require interpretation within their source models. In the WARM drywall calculation, the landfill result includes paper-facing carbon storage and an assumption of negligible methane yield. Recycling uses a modelled allocation of 19% to new drywall and 81% to agricultural uses [10]. EPA also identifies hydrogen sulphide risks under some disposal

conditions. A lower modelled GHG value therefore does not establish that a local disposal option is environmentally preferable or safe. Where one sentence in the EPA document conflicts with its numerical summary, the implemented factors follow the matching workbook cells E53/C53 and Exhibits 6-5/6-17 [10].

Material assignments must remain within the source categories. Cardboard is mapped to WARM Corrugated Containers, meaning cardboard with a corrugated inner layer. Folding cartons, mixed-material packages, bottle caps, and labels require checking rather than automatic assignment to these factors. Material identity, contamination, local rules, and facility acceptance also remain relevant. The comparisons do not verify actual treatment, establish emission reductions caused by the project, or demonstrate carbon-credit eligibility.

7. COST AND SCALABILITY ANALYSIS

7.1. Reported resources and cost boundary

The family reported a recurring hosting charge of IDR 600,000 per year. The laptop and scale were already owned, and no new hardware expenditure was reported. This amount covers the reported hosting charge only and should not be interpreted as the total project cost. Table 8 separates reported expenditure, existing equipment, and costs that have not been quantified.

Table 8. Reported expenditure, existing equipment, and unquantified costs as of 25 October 2026. Source: family-reported budget information and the implemented system design.

Resource	Reported amount / status	Cost Boundary
Shared hosting	IDR 600,000 per year	Family-reported recurring charge; not independently verified against an invoice.
Laptop	Already owned	No new purchase reported; acquisition cost and depreciation were not allocated to the project.
Scale	Already owned	No new purchase reported.
External prediction API	No external per-prediction service used	Inference runs locally; device electricity and computing-resource use remain unquantified.
Software / pretrained model	Existing open-source tools and pretrained artifacts	License conditions apply. Development effort is not included in the reported hosting charge.
Other costs	Not quantified	Development time, electricity, maintenance, equipment replacement, and any separate domain or bandwidth charges.

Image inference runs on the user's device without an external per-prediction AI service. Local processing still uses device electricity and computing resources. A complete budget would also account for equipment depreciation, development and data-preparation time, maintenance, scale checks, and eventual device replacement. Separate domain charges and bandwidth overages, where applicable, have not been quantified.

Energy consumption was not measured, and no comparison of total operating costs with alternative deployment approaches was performed. The reported budget therefore describes the prototype's known resources and expenditure without establishing energy savings or comparative cost-efficiency.

7.2. Scaling beyond one prototype

Scaling CarbonTrace involves serving application files to more users and maintaining reliable storage as records and uploaded photos accumulate. The evaluated static build contains approximately 26.6 MiB of files, including catalogue images and model artifacts. This is the aggregate uncompressed file size, not a measured first-visit download size. Shared hosting delivers the files, while image inference runs on each user's device. Performance therefore depends on both file delivery and the computing resources available on the client device.

Records and factor preferences are stored in local storage, while prepared copies of uploaded photos are stored in IndexedDB. These stores belong to a browser profile and website origin; they do not automatically synchronize across users or devices. Clearing site data can remove records and uploaded photos. The CSV export preserves selected record and provenance fields but contains no photo files and should not be treated as a complete, restorable application backup.

Larger deployments would require testing of browser storage limits, record-handling performance, and backup and recovery procedures. If school-wide use requires shared records or access across devices, additional infrastructure would be needed. It would also require access controls, consent and retention policies, and a way to combine records without losing identifiers, source snapshots, or the distinction between incompatible calculation bases.

Supporting another material would require an updated classifier and suitable labelled training and independently evaluated images across the supported classes. Emission estimates for that material would additionally require appropriate documented factors. Missing factors must remain unavailable rather than being copied from another material. Wider operational use would also require a tested approach to unsupported inputs and defined procedures for checking material identity and physical weights.

The present evaluation did not measure concurrent-user capacity, hosting performance under heavy traffic, or long-term growth of browser-held records. Actual recycling after recording was also not verified. These remain requirements for evaluating broader deployment and environmental outcomes.

8. CONCLUSIONS

CarbonTrace demonstrates a browser-based workflow connecting local AI material suggestions with human confirmation, entered weights, and saved emission-factor references. The study used one dataset of 120 photos covering PET bottles, cardboard, and paper-faced drywall, divided into 78 training photos and 42 evaluation photos. All 42 evaluation predictions matched their supplied labels, with 14 photos per class. All 146 software tests passed, including checks of record validation, factor-snapshot preservation, and scenario calculations. These results support the prototype's functionality and record traceability under the tested conditions.

The 1 kg scenario analysis showed that material and reference choices can change both the magnitude and direction of the estimated emission difference. Preserving these inputs and source assumptions allows users to inspect how a comparison was calculated. The results are analytical estimates, not observed emission reductions, and a lower modelled GHG value alone does not establish a safe or environmentally preferable treatment option. The limited internal image collection, similar backgrounds, unverified labels, and evaluation history also restrict conclusions about broader classification accuracy.

Further evaluation should use independently checked labels and new objects and scenes across all three classes. Real weighing and comparisons of recording with and without human review are needed to assess practical performance. Claims of environmental benefit would additionally require measured material quantities, evidence of actual treatment, and a suitable baseline. The

prototype provides a working basis for environmental learning and traceable scenario comparisons while these broader outcomes remain to be evaluated.

9. ACKNOWLEDGMENTS

The author thanks her family for project coordination, existing equipment, and budget information. The project also used the software and development support listed in the next section.

10. STUDENT ROLES

10.1. Student contribution

Rayna Amara Devini is the sole student author. Her contributions included selecting the problem, preparing and reviewing image labels, defining the recording workflow, examining the model results, and preparing the presentation. She is responsible for understanding the methods and explaining the findings and limitations.

10.2. External resources and contributions

The project uses pretrained MobileNet features, TensorFlow.js, and emission references from WARM and PLACES [2,3,5,6].

10.3. Data and reproducibility

The demonstration website provides the saved model report and model files [8]. The project archive retains the original PNG photographs, prepared JPEG copies, training settings, and test scripts. These materials and archived test output [9] can be supplied to support reproducibility. Private browser records are not part of the public research dataset. The reported findings cover image classification, software verification, and analytical emission-scenario calculations.

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